

OptOrch Local-Run Shiny App

A graphical user interface for
seed orchard contribution optimization

User Manual

Manuscript

OptOrch: A modular optimization toolkit for forest tree seed orchard deployment

Repository

<https://github.com/kyeji1107/OptOrch>

Date: 18.06.2026 | Version 1

R

SHINY

GUROBI

Quick-start workflow

Run the local Shiny app with your own input files.



Key requirement

The app runs in the user's local R environment and uses local Gurobi installation/license with BV.csv and coMatrix.csv as inputs.

Pre-run checklist

Confirm these items before launching the app.



Repository root

The working directory contains the **04_shiny_app** folder.



Input files

BV.csv and **coMatrix.csv** are stored in **04_shiny_app/Input_data**.



Matching IDs

IDs in **BV.csv** match the row and column names in **coMatrix.csv**.



Gurobi access

The gurobi R package can be loaded and the license is active.



Matrix scale

coMatrix.csv should contain a coancestry matrix. If you have a genetic or genomic relationship matrix, scale it by 0.5.

Install and check required R components

Install common R packages and confirm that Gurobi is available locally.

INSTALL PACKAGES

```
install.packages("shiny")  
install.packages("data.table")  
install.packages("Matrix")  
install.packages("ggplot2")
```

These packages support the Shiny interface, data handling, matrix operations, and plotting.

CHECK GUROBI

```
library(gurobi)
```

Gurobi must be installed separately. The R package should match the local Gurobi installation and license configuration.

Prepare the `Input\_data` folder

The app automatically looks for two CSV files inside `04\_shiny\_app/Input\_data`.

```
OptOrch/  
└─ 04_shiny_app/  
    └─ app.R  
        └─ Input_data/  
            └─ BV.csv  
                └─ coMatrix.csv
```

INPUT DATA CHECK

- BV.csv should contain individual IDs and breeding values.
- coMatrix.csv should contain a square pairwise coancestry matrix.
- If your source is a relationship matrix, multiply it by 0.5 before saving it as coMatrix.csv.
- When both files are found, the app shows “Ready to run: YES”.

Input file formats

Use simple CSV files and keep IDs consistent across both inputs.

BV.csv EXAMPLE

```
ID,BV
ind001,1.25
ind002,0.87
ind003,1.03
```

Required content: one ID column and one breeding value column. The app uses the individual IDs to align BV and coancestry information.

coMatrix.csv EXAMPLE

```
ID,ind001,ind002,ind003
ind001,0.50,0.02,0.03
ind002,0.02,0.50,0.01
ind003,0.03,0.01,0.50
```

Required content: a numeric, square, symmetric matrix. Row IDs and column IDs should match the IDs in BV.csv.

Launch the app from the R console

Set the working directory to the repository root, then call `shiny::runApp()`.

WINDOWS EXAMPLE

```
setwd("C:/Users/YourName/Downloads/OptOrch")  
shiny::runApp("04_shiny_app")
```

OPTIONAL CHECKS

```
getwd()  
list.files()  
list.files("04_shiny_app/Input_data")
```

MAC/LINUX EXAMPLE

```
setwd("~/Downloads/OptOrch")  
shiny::runApp("04_shiny_app")
```

Before launching

- The working directory should contain `04_shiny_app`.
- `BV.csv` and `coMatrix.csv` should be in `Input_data`.
- Gurobi R package and license should be configured locally.

Interface overview

OptOrch Local-Run Shiny App

1

Input data check

Input data folder: FOUND
BV.csv: FOUND
coMatrix.csv: FOUND
Ready to run: YES

Refresh input data status

Scenario

Scenario 1: Status number

Status number (Ns)

10

Gurobi parameter setting

Default

Time limit

None

MIPGap (optional)

Leave blank to use Gurobi default (0.0001); e

Lower contribution bound

0.01

Upper contribution bound

0.15

☒ Print Gurobi log in R console while running

Run optimization

Download selected individuals

2

3

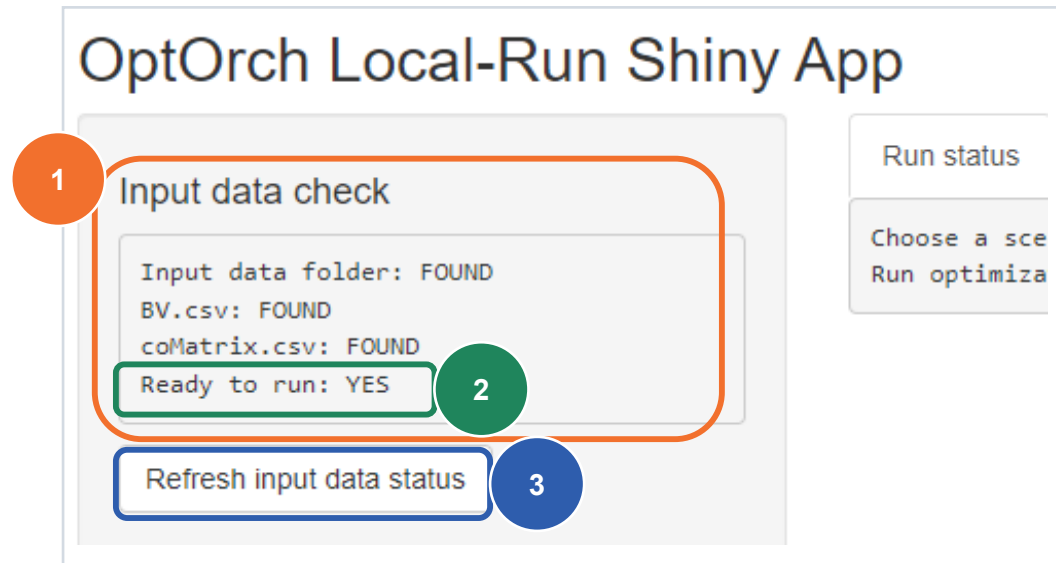
Run status | Summary | Selected individuals | Contribution plot | Gurobi log

Choose a scenario and click 'Run optimization'.
Run optimization uses BV.csv and coMatrix.csv in 04_shiny_app/Input_data.

Where to look first

- 1 Left sidebar**
Inputs, scenarios, solver settings, contribution bounds, run button, and download button.
- 2 Result area**
Run status and result tabs are displayed on the right side of the app window.
- 3 Navigation tabs**
Use the tabs to move between Summary, Selected individuals, Contribution plot, and Gurobi log.

Input data check



Confirm input files before running

- 1 Input data status**
The app checks whether Input_data, BV.csv, and coMatrix.csv are found.
- 2 Ready to run**
When all inputs are found, the message shows Ready to run: YES.
- 3 Refresh button**
Click this after replacing or adding input files in the Input_data folder.

If Ready to run is NO, check file names and the 04_shiny_app/Input_data folder.

Scenario and Status number controls

OptOrch Local-Run Shiny App

Input data check

Input data folder: FOUND
BV.csv: FOUND
coMatrix.csv: FOUND
Ready to run: YES

Refresh input data status

Run status

Choose a scen
Run optimizat

Scenario

Scenario 1: Status number

Status number (Ns)

10

Gurobi parameter setting

Default

Time limit

None

Set the main optimization scenario

- 1 Scenario dropdown**
Choose Scenario 1: Status number or Scenario 2: Pollen contamination.
- 2 Status number (Ns)**
Ns is used in the diversity constraint; it does not force the app to select exactly Ns individuals.
- 3 Parameter preset**
Use Default or a tuned parameter set. The tuned parameter sets are the Gurobi settings reported in Table 1 of the manuscript. They can be optionally applied according to the selected status number level.

Default applies only the basic settings:
OutputFlag = 1 and NonConvex = 2.

Gurobi parameter settings

The screenshot shows a web-based interface for Gurobi parameter settings. It includes several input fields and a checkbox, each with a numbered callout:

- Status number (Ns)**: A text input field containing the value "10".
- Gurobi parameter setting**: A dropdown menu currently set to "Default".
- Time limit**: A dropdown menu set to "None", highlighted with an orange box and callout 1.
- MIPGap (optional)**: A text input field containing the text "Leave blank to use Gurobi default (0.0001); e", highlighted with a blue box and callout 2.
- Lower contribution bound**: A text input field containing the value "0.01".
- Upper contribution bound**: A text input field containing the value "0.15".
- Print Gurobi log in R console while running**: A checkbox that is checked, highlighted with a green box and callout 3.

At the bottom of the form are two buttons: "Run optimization" and "Download selected individuals".

Control solver behavior

- 1 Time limit**
Choose None or a fixed time limit. If reached, the app reports TIME_LIMIT.
- 2 MIPGap**
Leave blank for the Gurobi default, or enter a numeric value for a custom tolerance.
- 3 Console log option**
Enable this to print the Gurobi log in the R console while the model is running.

For quick testing, a larger MIPGap can reduce runtime; stricter optimization uses smaller values.

Contribution bounds and run controls

The screenshot shows a web-based interface for Gurobi parameter settings. It includes several sections: 'Gurobi parameter setting' with a 'Default' dropdown, 'Time limit' with a 'None' dropdown, and 'MIPGap (optional)' with a text input containing 'Leave blank to use Gurobi default (0.0001); ε'. Below these are two input fields for contribution bounds: 'Lower contribution bound' with the value '0.01' (highlighted with an orange box and callout 1) and 'Upper contribution bound' with the value '0.15' (highlighted with a blue box and callout 2). A checkbox labeled 'Print Gurobi log in R console while running' is checked. At the bottom are two buttons: 'Run optimization' (highlighted with a teal box and callout 3) and 'Download selected individuals' (highlighted with a green box and callout 4).

Run optimization and export results

- 1 Lower contribution bound**
Minimum contribution assigned to each selected individual. Example: 0.01.
- 2 Upper contribution bound**
Maximum contribution allowed for one selected individual. Example: 0.15.
- 3 Run optimization**
Starts the Gurobi optimization using BV.csv and coMatrix.csv.
- 4 Download selected individuals**
Exports selected ID, BV, optimized contribution p, and binary variable r.

Scenario 2 pollen contamination inputs

Lower contribution bound

0.01

Upper contribution bound

0.15

Pollen contamination rate

0.3

Number of external pollen parents

100

Mean BV of external pollen parents

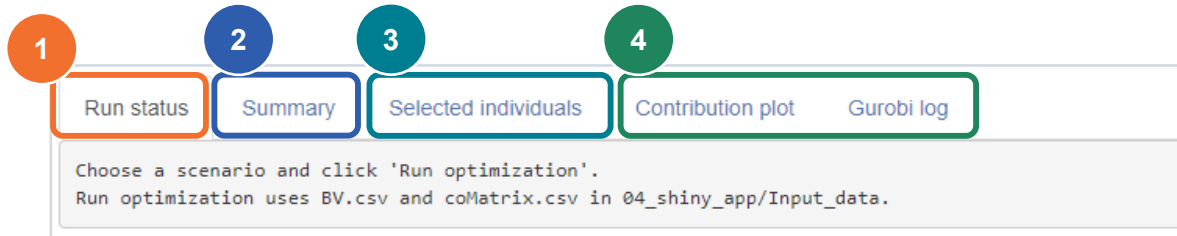
0

'Scenario 2'-specific controls

- 1 Pollen contamination rate**
Proportion of pollen originating from outside the seed orchard. Example: 0.30
- 2 Number of external pollen parents**
Number of pollen contaminants from outside seed orchard. Example: 100
- 3 Mean BV of external pollen parents**
User-specified mean breeding value of pollen contaminants used when calculating total gain.

These inputs appear only when 'Scenario 2: Pollen contamination' is selected.

Result tabs



Review the optimization output

- 1 Run status**
Shows the current solver status and short instructions before or after the run.
- 2 Summary**
Reports objective value, runtime, status, and number of selected individuals.
- 3 Selected individuals**
Lists selected IDs, BV values, optimized contributions p , and selection variable r .
- 4 Plot and log**
Contribution plot visualizes p ; Gurobi log supports troubleshooting.

Start with Run status and Summary, then inspect Selected individuals and Gurobi log if needed.

Run status messages

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☒ Print Gurobi log in R console while running

Run optimization

Download selected individuals

Run status

Summary

Selected individuals

Contribution plot

Gurobi log

Choose a scenario and click 'Run optimization'.
Run optimization uses BV.csv and coMatrix.csv in 04_shiny_app/Input_data.

Interpret solver output

- Before run**
The Run status tab asks you to choose a scenario and click Run optimization.
- 'OPTIMAL'**
A solution satisfying the selected optimality tolerance was found.
- 'INFEASIBLE'**
No solution satisfies all constraints. Check Ns, bounds, and coMatrix scaling.
- 'TIME_LIMIT'**
The selected time limit was reached. A feasible solution may still be available.

For unexpected results, review both Summary and Gurobi log.